Sachin Bhadang

+1 (619) 960-7982 | sbhadang@ucsd.edu | sachinb20.github.io

Education

University of California San Diego (UCSD) **Sep 2025 – Jun 2027**
Master of Science in ECE – Intelligent Systems, Robotics and Control *San Diego, CA*

Indian Institute of Technology (IIT) Kanpur **Jul 2020 – May 2025**
Bachelor of Technology in Mechanical & Electrical Engineering (Double Major) *Kanpur, India*

Experience

Wang Lab, UC San Diego **Jan 2026 – Present**
Graduate Researcher (Advisor: Prof. Xiaolong Wang) *San Diego, CA*

- Developed a cross-modal framework to align **resistive, magnetic, and vision-based** tactile sensors in a shared latent space using a **CVAE** with cross-reconstruction and contrastive losses.
- Trained and evaluated tactile-conditioned policies for **plug insertion**, **board wiping**, **pick-and-place**, and **object reorientation** to enable **zero-shot deployment** across different tactile sensors without retraining.
- Conducted **repeatability and stress tests** to characterize tactile sensor drift and evaluated the effects of **normal and shear force information** on downstream manipulation policy performance.
- Developed **force-aware retargeting** for dexterous hands and contributed to ongoing **dexterous-hand benchmarking**.

Kikobot **Jun 2025 – Sep 2025**
Firmware/Robotics Engineer *Pune, India*

- Implemented custom **field-oriented control (FOC)** firmware on **STM32 MCUs** for Harmonic Drive BLDC motors; integrated TI driver boards and **MA600 encoders over SPI**, designed a **CAN-FD** bus for multi-joint coordination, and validated gravity compensation to improve backdrivability and human-robot safety.
- Integrated a **6-DOF manipulator** with **Hugging Face LeRobot** for data collection and imitation learning; trained **ACT and diffusion policies**, applied **novel-view synthesis** to mitigate camera jitter, and developed an embedded **inverse-kinematics solver** without ROS and an operator UI for Windows and Linux.

PEARL Lab, TU Darmstadt | Code 1 | Code 2 | Report **Jan 2024 – Aug 2024**
Research Intern (Advisor: Prof. Georgia Chalvatzaki) *Darmstadt, Germany*

- Designed **action-conditioned scene graphs** and trained **CNN-GRU PPO policies** in **AI2-Thor** to reveal occluded and hidden objects in interactive environments.
- Built a **replanning pipeline** that used visual feedback to detect subtask failures and environment changes and support dynamic replanning in long-horizon manipulation tasks.
- Developed exploration policies for hierarchical **scene representations** of object-receptacle relationships and validated the methods in simulation and on the **Tiago++ mobile manipulator**.

Projects

SmolVAM: Lightweight Video-Action Models for Robot Control **(Code) | (Report)**

- Developed a lightweight **video-action model (VAM)** for robotic manipulation by combining a **0.7B Hummingbird video model**, **VideoMAE** spatiotemporal representations, and a transformer-based **flow-matching** action decoder.
- Fine-tuned the generative world model with **LoRA** on **20 MimicGen task variations**, including stacking, threading, peg insertion, and coffee preparation, and achieved a **60% success rate** on robotic cube-stacking tasks.

IMU Orientation Tracking & Panorama Reconstruction **(Code) | (Report)**

- Implemented **IMU calibration** and **quaternion-based orientation tracking** with projected and geodesic gradient descent to combine gyroscope motion models with accelerometer gravity observations.
- Reduced mean optimization iterations from **948 to 343 (∼64%)** across **nine training datasets** using motion-model initialization instead of identity initialization.
- Reconstructed **spherical panoramas** using synchronized camera frames, estimated orientations, and **equirectangular projection**, and compared alpha-blending and overwrite strategies.

Publications & Patents

Published / Accepted Papers & Granted Patents

TACTX: Learning Shared Tactile Representations Across Diverse Sensors
Junsung Park*, **Sachin Bhadang***, Carmelo Sferrazza, Sha Yi, Xiaolong Wang.
Conference on Robot Learning (CoRL), 2026. (arXiv)

*Equal contribution.

Method and System of Brainwave Controlled Wheelchair with Intelligent Tagging Assistant
J. Meenakshinathan, T. Sandhan, K. Gupta, **S. Bhadang**, Laxmidhar Behera.
Indian Patent No. 534723; granted Apr 24, 2024. (Patent Details)

Application No. 202311000340; filed Jan 3, 2023.

Bin-picking Novel Objects through Category-Agnostic Segmentation: RGB Matters
P. Raj, **S. Bhadang**, G. Chaudhary, T. Sand, L. Behera.
IEEE International Conference on Robotic Computing (IRC), 2023. (arXiv)

Learning 6DoF Grasp-poses in Occluded Scenes Using RGB and 3D Point Cloud Modalities
A.D. Mathur, **S. Bhadang**, P. Raj, L. Behera, T. Sandhan.
IEEE National Conference on Communications (NCC), 2023. (DOI)

Under Review

RetargetingKit: A Modular Toolkit for Dexterous Retargeting Beyond Kinematics
Changwei Jing, Xuanbin Peng, **Sachin Bhadang**, Yuhang Wu, Lingqi Zeng, Dongjun Tan, Keyi Wang, Xiaolong Wang, Jianglong Ye.
IEEE International Conference on Robotics and Automation (ICRA), 2027 — Under review.

TacO: Benchmarking Tactile Sensors for Object Manipulation
Anya Zorin, Zilin Si, **Sachin Bhadang**, Myungsun Park, Junsung Park, Alexiy Buynitsky, Taejun Park, Sohee John Yoon, Yong-Lae Park, Oliver Kroemer, Zeynep Temel, Michael T. Tolley, Sha Yi, Xiaolong Wang.
IEEE International Conference on Robotics and Automation (ICRA), 2027 — Under review. (arXiv)

Technical Skills

Programming: C, C++, Python, Bash
ML & Robotics Software: PyTorch, Open3D, PyQt, ROS, RViz
Simulation: Isaac Lab, Gazebo, MuJoCo, PyBullet, AI2-Thor, Habitat
Development & Computing Tools: Git, Bazel, SLURM, LaTeX
Embedded Systems & Motor Control: STM32, Arduino, FOC, CAN-FD, SPI
CAD & PCB Design: Onshape, Fusion 360, Eagle, EasyEDA
Robots & Dexterous Hands: Franka Panda, UR5/UR10, ClearPath Husky, Tiago++, Unitree Go2, Sharpa Wave, XHand

Selected Achievements

Dec 2023: One of three IIT Kanpur students nominated for an exchange at **EPFL, Switzerland**.
Dec 2022: Selected for a **Mitacs Globalink Research Internship** for fully funded research in Canada.
Jun 2022: One of the top 10 students in a cohort of 1,200 offered a **double major in Electrical Engineering**.
May 2022: Third place among 40 international teams in the **IEEE ICRA DJI RoboMaster AI Challenge**.
Dec 2021: Special Mention for innovative mechanical design at the Inter-IIT **AIITRA Robotics Challenge**.
Sep 2020: Ranked in the **top 1.3%** among 160,000 candidates in **JEE Advanced**, India.